

IRIS–Eventide Initiative: Relativistic Propulsion via Controlled Micro Black Hole Systems

Abstract—Current propulsion technologies based on chemical and nuclear processes are fundamentally constrained by limited specific energy and exponential mass penalties, preventing practical interstellar travel within human timescales. This work explores a conceptual propulsion architecture that departs from conventional reaction-based systems and instead leverages relativistic physics through controlled mass–energy conversion.

We propose the IRIS (Interstellar Relativistic Initiative System) under the Eventide Program, a theoretical framework for propulsion based on the creation, stabilization, and utilization of micro black hole systems as high-efficiency energy converters. By exploiting Hawking evaporation, such systems can, in principle, transform infalling mass into directed high-energy radiation with near-unity efficiency.

A generalized dynamic model is developed in which the black hole mass is treated as a time-dependent variable subject to competing processes: Hawking evaporation and externally supplied mass flux. Under equilibrium conditions, a steady-state regime emerges, enabling continuous power output. We examine the conditions required for this regime and derive scaling relations for power, thrust, and system lifetime.

Key engineering challenges are identified, including: (1) generation of sufficient energy density to produce a micro black hole (e.g., via a Kugelblitz mechanism), (2) efficient mass injection under extreme gravitational cross-section constraints, (3) conversion and collimation of high-energy isotropic radiation into directed thrust, and (4) dynamic stabilization of the black hole relative to the spacecraft without reliance on electrostatic confinement.

Order-of-magnitude analysis suggests that, if these challenges are overcome, such ulsion systems could achieve relativistic exhaust characteristics and sustained high thrust-to-mass ratios, potentially enabling interstellar transit to nearby systems such as Alpha Centauri within mission-relevant timescales.

This study does not claim near-term feasibility; rather, it establishes a physically consistent framework for future investigation into propulsion systems operating at the interface of general relativity and high-energy physics. The IRIS–Eventide Initiative is positioned as a long-horizon research program aimed at redefining propulsion beyond the classical rocket paradigm.

I. INTRODUCTION

The pursuit of interstellar travel remains one of the most demanding challenges in modern aerospace engineering and applied physics. Despite over a century of progress in propulsion systems, current technologies remain fundamentally constrained by limitations in achievable specific energy and propulsion efficiency. Chemical propulsion, which dominates present-day launch and in-space maneuvering architectures, is bounded by molecular bond energies and is therefore incapable of supporting sustained relativistic travel. Even advanced nuclear concepts—such as fission- and fusion-based propulsion—while offering several orders of magnitude improvement in energy density, remain insufficient for rapid interstellar

transit when evaluated under the constraints imposed by the Tsiolkovski Rocket Equation.

These limitations have motivated the exploration of propulsion paradigms that extend beyond classical reaction-based systems. Prior studies have investigated concepts including beamed energy propulsion, antimatter-based drives, and externally powered sail architectures. While these approaches partially mitigate the mass constraints of onboard propellant, they introduce new dependencies on large-scale infrastructure, limited thrust regimes, or unresolved challenges in energy production and storage.

In parallel, developments in high-energy physics and gravitation theory have suggested that more fundamental approaches to propulsion may be possible by directly exploiting relativistic phenomena. In particular, the theoretical framework of the General Relativity admits the existence of black holes as compact objects capable of converting mass into radiation through quantum effects. The process of Hawking evaporation provides a mechanism by which black holes emit energy with an efficiency approaching unity relative to the rest mass of infalling matter, representing the most energy-dense conversion process currently described by known physics.

The concept of artificially generating black holes via concentrated energy—commonly referred to as a Kugelblitz—has been discussed in theoretical literature as a potential pathway to accessing such energy conversion regimes. Although no experimental realization currently exists, the underlying principles are consistent with established relativistic and quantum field frameworks. If achievable, such systems could serve as compact, high-power sources capable of sustaining propulsion architectures far exceeding the performance limits of conventional engines.

This work introduces the IRIS (Interstellar Relativistic Initiative System) under the Eventide Program, a conceptual propulsion framework based on controlled micro black hole systems. The approach departs from traditional rocket architectures by treating the black hole not as a passive gravitational object, but as an active mass–energy transducer within a closed dynamic system. By coupling Hawking radiation with controlled mass injection, it is theoretically possible to establish a steady-state regime in which the black hole maintains a quasi-constant mass while continuously emitting high-energy radiation.

The objective of this study is not to demonstrate near-term feasibility, but to define a physically consistent and mathematically tractable framework for such systems. We aim to (1) establish the governing equations for mass evolution

and energy output, (2) evaluate the conditions required for equilibrium operation, (3) identify the dominant physical and engineering constraints, and (4) assess the implications of such systems for high-thrust, high-efficiency propulsion. Particular attention is given to the challenges of radiation management, momentum transfer, and system stability, which represent critical barriers to implementation.

By situating the IRIS–Eventide concept within the broader context of propulsion research, this work seeks to contribute to long-horizon investigations into technologies that may ultimately enable practical interstellar exploration. While the technological requirements are far beyond current capabilities, the underlying physics remains grounded in established theory, providing a basis for continued analytical and computational study.

II. PHYSICAL MODEL

The IRIS–Eventide propulsion concept is based on the controlled interaction between a micro black hole and its surrounding engineered environment. The system is modeled as a dynamic mass–energy converter, in which the black hole mass evolves under the competing effects of quantum evaporation and externally supplied mass flux.

A. Black Hole Fundamental Properties

We consider a non-rotating, uncharged black hole described by the Schwarzschild metric. Its characteristic radius is given by:

$$R_s = \frac{2GM}{c^2} \quad (1)$$

where G is the gravitational constant, M is the black hole mass, and c is the speed of light. The black hole emits radiation via quantum effects near the event horizon, with total power output described by:

$$P(M) = \frac{\hbar c^6}{15360\pi G^2 M^2} \quad (2)$$

This relation implies an inverse-square dependence of radiated power on mass, leading to dramatically increasing power output as the black hole mass decreases.

B. Mass Evolution Dynamics

The time evolution of the black hole mass is governed by the balance between Hawking evaporation and external mass injection:

$$\frac{dM}{dt} = \dot{M}_{in} - \frac{P(M)}{c^2} \quad (3)$$

Substituting the expression for $P(M)$:

$$\frac{dM}{dt} = \dot{M}_{in} - \frac{\hbar c^4}{15360\pi G^2 M^2} \quad (4)$$

This nonlinear differential equation defines the core dynamics of the system.

C. Equilibrium Regime

A steady-state operating condition is achieved when:

$$\frac{dM}{dt} = 0 \quad (5)$$

which yields:

$$\dot{M}_{in} = \frac{P(M)}{c^2} \quad (6)$$

Under this condition, the black hole maintains constant mass while continuously converting infalling matter into radiation. This regime represents the ideal operational mode for propulsion, enabling sustained power output without runaway evaporation or uncontrolled growth.

D. Thrust Generation

To generate propulsion, a fraction $\eta \in [0, 1]$ of the emitted radiation must be directionally converted into momentum flux. The resulting thrust is given by:

$$F = \frac{\eta P(M)}{c} \quad (7)$$

where η represents the efficiency of radiation collimation and conversion. Substituting for $P(M)$:

$$F = \frac{\eta \hbar c^5}{15360\pi G^2 M^2} \quad (8)$$

This expression highlights a critical design trade-off:

- smaller $M \rightarrow$ higher thrust
- smaller $M \rightarrow$ lower stability and shorter lifetime

E. System Stability

The equilibrium condition is dynamically sensitive. Perturbations in mass lead to asymmetric responses:

- If M decreases:
 - $P(M) \uparrow$
 - evaporation dominates $\rightarrow$ runaway collapse
- If M increases:
 - $P(M) \downarrow$
 - accumulation dominates $\rightarrow$ growth

Thus, the equilibrium is intrinsically unstable and requires active feedback control through precise modulation of $\dot{M}_{in}$.

F. Characteristic Scaling Laws

From the governing equations, several key scaling relationships emerge:

- Power: $P \propto M^{-2}$
- Thrust: $F \propto M^{-2}$
- Lifetime (unfed): $\tau \propto M^3$

These relations define the design space of the propulsion system and constrain feasible operating regimes.

G. System Interpretation

The IRIS–Eventide system can be interpreted as a relativistic engine in which mass is continuously converted into directed energy with near-maximal efficiency, unlike conventional propulsion:

- no large propellant mass is required
- energy density is not chemically limited
- thrust is decoupled from traditional exhaust mechanisms

Instead, performance is governed by:

- black hole mass
- energy conservation efficiency
- control of relativistic processes

H. Section Summary

This model establishes a self-consistent physical framework for a propulsion system based on micro black hole dynamics. While idealized, it captures the essential behavior required to evaluate feasibility, performance limits, and control requirements.

III. SYSTEM ARCHITECTURE

The IRIS–Eventide propulsion system is conceived as an integrated relativistic engine in which a micro black hole is maintained in a controlled environment and coupled to subsystems responsible for mass injection, energy conversion, thrust generation, and stabilization. The architecture departs fundamentally from conventional spacecraft design, requiring axial symmetry, extreme energy management, and dynamic control at relativistic scales.

A. Global Configuration

The spacecraft is organized along a primary longitudinal axis, with the propulsion system located at its core. The micro black hole is positioned within a central axial cavity (hereafter referred to as the reaction channel), extending through the length of the vehicle. This configuration provides:

- geometric isolation between the black hole and critical systems
- a natural direction for thrust generation
- reduced structural exposure to radiation and tidal gradients

The spacecraft can be **conceptually** divided into three main sections:

- 1) Forward Section
 - payload, control systems, crew (if applicable)
 - shielded from direct radiation exposure
- 2) Central Section (Reaction Channel)
 - black hole containment region
 - mass injection systems
 - radiation conversion structures
- 3) Aft Section
 - exhaust direction
 - energy redirection and thrust vectoring

B. Black Hole Positioning and Reaction Channel

The micro black hole is maintained near the geometric center of the reaction channel but offset toward the aft direction to bias momentum flow. Rather than physical containment, the system relies on:

- dynamic equilibrium of incoming particle fluxes
- symmetric or controlled-asymmetric injection geometries

The channel itself is a hollow, cylindrical structure designed to:

- allow unobstructed propagation of high-energy radiation
- minimize interaction with structural materials
- define the thrust axis

This directly reflects your earlier intuition of a “tunnel” geometry—formalized here as a necessary architectural feature, not an optional one.

C. Mass Injection System

The mass injection subsystem is responsible for sustaining the equilibrium condition:

$$\dot{M}_{in} = \frac{P(M)}{c^2} \quad (9)$$

It consists of:

- high-energy particle accelerators
- beam focusing and steering systems
- injection nozzles arranged radially or axially

Key design principles:

- Relativistic injection velocities to maximize capture probability
- Angular momentum control to avoid stable orbits
- multi-directional injection for stabilization

Rather than discrete “particles one by one,” the system operates as a continuous relativistic mass flux, tuned in real time based on feedback from black hole emission measurements.

D. Radiation Conversion and Thrust Generation

The black hole emits nearly isotropic high-energy radiation. To produce thrust, this radiation must be partially converted into a directed momentum flux. Instead of relying on passive shielding, the system employs an active conversion layer: Possible mechanisms (theoretical):

- pair production and magnetic channeling
- absorption in dense plasma and re-emission
- relativistic particle redirection

The aft section of the reaction channel acts as an effective relativistic exhaust nozzle, where:

- forward-directed radiation is absorbed or redirected
- aft-directed radiation is allowed to escape or is enhanced

This creates a net momentum flux:

$$F = \frac{\eta P(M)}{c} \quad (10)$$

where η depends entirely on the effectiveness of this subsystem.

E. Radiation Shielding and Structural Protection

Given the extreme radiation environment, shielding is not treated as a passive barrier but as a multi-layer defense system:

- 1) Geometric shielding
 - axial separation
 - shadowing of critical systems
- 2) Material shielding
 - dense materials for partial attenuation
 - sacrificial layers
- 3) Field-based mitigation (theoretical)
 - magnetic deflection of charged particles

The forward section is positioned outside the primary radiation cone, reducing exposure through geometry rather than mass alone.

F. Dynamic Stabilization

The black hole cannot be mechanically supported and must be stabilized dynamically. Primary mechanisms:

- 1) Momentum balancing via injection
 - adjusting direction and intensity of incoming mass flux
 - effectively “pushing” the black hole into position
- 2) Symmetry control
 - multi-beam configurations maintaining equilibrium
- 3) Feedback loop
 - real-time monitoring of:
 - radiation output
 - inferred position
 - mass variation

This forms a closed control system analogous to a relativistic-scale inverted pendulum problem.

G. Control System

The propulsion system requires continuous feedback and adjustment:

- sensors measuring radiation flux and spectrum
- estimation of black hole mass and position
- control algorithms regulating $\dot{M}_{in}$ and beam geometry

Control operates on extremely short timescales due to the nonlinear nature of the system.

H. System-Level Interpretation

The complete architecture can be understood as a self-regulated relativistic engine in which a micro black hole acts as a continuously fed energy conversion core, coupled to a directed radiation exhaust system. Unlike classical propulsion systems:

- there is no discrete combustion chamber
- no traditional exhaust plume
- no reliance on stored chemical energy

Instead, the system operates as a mass–energy throughput device, where the input is the controlled mass flux and the output is directed relativistic energy.

I. Section Summary

The proposed architecture translates the theoretical model into a coherent spacecraft design framework. While highly speculative, each subsystem corresponds to a defined physical requirement derived from the governing equations.

IV. ENGINEERING CHALLENGES AND CONSTRAINTS

While the IRIS–Eventide architecture is consistent with established physical principles, its realization is constrained by multiple extreme engineering challenges. These challenges arise from the unprecedented energy scales, precision requirements, and physical regimes involved. This section identifies the dominant barriers and frames them as research directions rather than immediate implementation targets.

A. Energy Density and Black Hole Formation

The creation of a micro black hole via a Kugelblitz mechanism requires concentrating energy on the order of:

$$E \sim Mc^2 \quad (11)$$

into a region comparable to its Schwarzschild radius. For even modest black hole masses, this implies:

- energy levels far exceeding current global production capabilities
- spatial confinement at subatomic scales
- synchronization of energy delivery at relativistic precision

The primary limitation is not total energy generation, but energy density and focusing, which remain many orders of magnitude beyond current technology.

B. Mass Injection Efficiency

Sustaining the equilibrium condition requires continuous and precise mass delivery into an extremely small effective capture cross-section. Key challenges include:

- low probability of particle capture due to small Schwarzschild radius
- scattering and deflection of incoming particles
- sensitivity to angular momentum leading to non-accreting trajectories

Even with advanced accelerator systems, achieving efficient coupling between injected matter and the black hole remains an unresolved problem in relativistic particle dynamics.

C. Radiation Conversion and Collimation

The emitted Hawking radiation is expected to be:

- broadband
- highly energetic (gamma-dominated for small masses)
- nearly isotropic

Transforming this emission into directed thrust requires:

- interaction with matter or fields under extreme conditions
- conversion mechanisms that preserve momentum efficiency
- materials or plasma states capable of surviving high energy fluxes

No known material can directly reflect or contain such radiation, making indirect conversion mechanisms a necessity.

D. Thermal and Structural Constraints

The energy flux associated with micro black hole emission can exceed any currently manageable thermal load. Consequences include:

- rapid material degradation
- plasma formation and structural ablation
- difficulty maintaining stable geometries

This necessitates:

- non-traditional structural concepts (e.g., sacrificial or self-regenerating layers)
- reliance on geometry and distance rather than mass shielding alone

E. Dynamic Stability and Control

As established in Section 2, the system equilibrium is intrinsically unstable. Small perturbations in mass lead to:

- runaway evaporation
- uncontrolled growth

Maintaining stability requires:

- real-time measurement of system state variables
- ultra-fast control of mass injection rates
- precise spatial targeting of relativistic particle beams

This introduces a control problem operating at the limits of measurement resolution and response time.

F. Black Hole Positioning and Containment

The black hole cannot be physically supported or enclosed. Challenges include:

- preventing drift toward structural components
- avoiding gravitational interaction with critical systems
- maintaining alignment with the thrust axis

Electromagnetic confinement is ineffective unless the black hole retains significant charge, which is expected to dissipate rapidly. Therefore, stabilization must rely on dynamic momentum balancing, increasing system complexity.

G. Radiation Hazard and System Survivability

The radiation environment poses severe risks:

- ionizing damage to electronics
- material embrittlement
- secondary particle cascades

Even indirect exposure may compromise system integrity over time. Shielding strategies must therefore combine:

- geometric isolation
- active field-based mitigation (where applicable)
- system redundancy

H. Scaling and Mission Integration

The performance advantages of the system depend strongly on black hole mass, however:

- smaller masses → higher power but lower controllability
- larger masses → more stable but lower thrust

This creates a constrained design space in which:

- mission duration

- acceleration profile
- system mass

must be jointly optimized.

I. Section Summary

The IRIS–Eventide propulsion system faces challenges that extend beyond current engineering capabilities, particularly in the domains of energy concentration, radiation management, and dynamic control. These constraints do not invalidate the underlying concept, but they define a set of critical research frontiers that must be addressed before practical implementation can be considered.

V. DISCUSSION AND FUTURE WORK

The IRIS–Eventide Initiative presents a conceptual propulsion framework that operates at the intersection of high-energy physics and relativistic gravitation. By modeling micro black holes as controllable mass–energy transducers, this work outlines a pathway—albeit highly speculative—toward propulsion systems capable of exceeding the fundamental limitations of chemical and conventional nuclear technologies. In comparison with existing advanced propulsion concepts, the proposed architecture offers several theoretical advantages. Unlike systems constrained by the Tsiolkovski Rocket Equation, the IRIS–Eventide model decouples propulsion performance from onboard propellant mass in the traditional sense, instead relying on continuous mass–energy conversion. Relative to beamed propulsion or sail-based systems, it eliminates dependence on external infrastructure once initiated. Furthermore, when compared to antimatter-based propulsion, it provides a conceptually similar energy density regime without requiring the storage of large quantities of unstable fuel, though at the cost of significantly greater complexity in system realization. However, the analysis presented here also makes clear that the gap between theoretical consistency and engineering feasibility remains substantial. The challenges identified—particularly in energy concentration, radiation conversion, and dynamic stabilization—are not incremental extensions of current technology, but rather require advances across multiple domains of physics and engineering. As such, the IRIS–Eventide Initiative should be interpreted not as a near-term development program, but as a long-horizon research framework. Within this context, several priority areas for future work can be identified:

A. Theoretical and Computational Development

- refinement of black hole mass evolution models under realistic injection conditions
- numerical simulations of particle capture efficiency in relativistic regimes
- modeling of radiation–matter interactions at extreme energy densities

These efforts will help determine whether stable operating regimes can be sustained under physically plausible conditions.

B. Energy Generation and Concentration

- investigation of high-density energy accumulation mechanisms
- exploration of coherent energy delivery systems capable of extreme spatial focusing
- evaluation of fusion-based or alternative energy sources as precursors

Advances in this area are prerequisite for any attempt at black hole generation, including Kugelblitz scenarios. This represents one of the most critical unresolved challenges in translating energy output into usable thrust.

C. Radiation Conversion Mechanisms

- study of indirect collimation techniques using plasma or field-mediated processes
- exploration of pair-production cascades and their controllability
- development of models for high-efficiency momentum transfer from isotropic radiation

This represents one of the most critical unresolved challenges in translating energy output into usable thrust.

D. Control Systems and Stability

- design of feedback systems capable of regulating mass injection at relativistic timescales
- investigation of multi-beam stabilization strategies
- analysis of system response to perturbations and failure modes

The intrinsic instability of the equilibrium regime requires control approaches beyond current aerospace standards.

E. Experimental Proxies

Given the infeasibility of direct experimentation, intermediate steps may include:

- high-energy plasma confinement experiments
- particle beam interaction studies at extreme densities
- analog systems for testing feedback and control architectures

Such efforts could provide partial validation of subsystem behaviors.

F. Strategic Context

Research into propulsion systems of this class aligns with long-term goals of expanding human presence beyond the Solar System. While current programs led by organizations such as NASA and other international agencies focus primarily on near-term and mid-term exploration capabilities, there is increasing recognition of the need for foundational studies addressing interstellar travel. The IRIS–Eventide Initiative is positioned within this broader context as an exploratory effort aimed at defining the boundaries of physically permissible propulsion. Its value lies not in immediate application, but in contributing to the conceptual and analytical groundwork upon which future technologies may be built.

VI. CONCLUSION

This study has outlined a physically consistent, though highly speculative, propulsion concept based on controlled micro black hole systems. By integrating principles from general relativity, quantum field theory, and high-energy engineering, it establishes a framework for further investigation into propulsion mechanisms operating far beyond current technological limits. The transition from concept to realization—if possible—will require sustained advances across multiple scientific domains. Nevertheless, the exploration of such ideas plays a critical role in expanding the scope of aerospace research and defining long-term trajectories for technological development.